

and can be installed easily, according to requirements.

- Offers caching and offline analysis/replaying of test results.

The cons of JMeter

- It can be used only on Web applications.
- Consumption of memory is high in GUI mode, and load, stress and endurance testing with high user loads should preferably be run in non-GUI mode.
- Complex scenarios cannot be checked using JMeter thread group.
- Recording is complex, as we need to set up the proxy manually.
- It supports only Java for custom coding.

2. Capybara: Acceptance test framework for Web applications

Capybara is a Web based automation framework used for creating functional tests that simulate how users interact with your application. It is a library built to be used on top of an underlying Web based driver. It offers a user friendly DSL (domain specific language), which is used to describe actions that are executed by the underlying Web driver. When the page is loaded using the DSL (and underlying Web driver), Capybara will try to locate the relevant element in the DOM (Document Object Model) and execute the action, such as click a button, link, etc.

The pros of Capybara

- No set-up necessary for Rails and Rack applications. It works out-of-the-box.
- Intuitive API, which mimics the language an actual user would use.
- Powerful synchronisation features mean you never have to manually wait for asynchronous processes to complete.



Figure 2: Selenium's capabilities

- Capybara uses the same DSL to drive a variety of browsers and headless drivers.

The cons of Capybara

- The only con of this tool is that its framework adds a layer on top of the actual implementation which makes it tough to debug what is actually happening.

3. Selenium: Web app testing tool

Selenium is a suite of tools such as Selenium IDE, Selenium Remote Control and Selenium Grid to test the Web application. Selenium IDE is an integrated development environment for Selenium scripts. It is implemented as a Firefox extension, and allows you to record, edit, and debug tests. It supports record and playback.

The pros of Selenium

- It is a low cost tool.
- It can carry out browser compatibility testing.
- It offers a choice of languages.
- It has multiple testing frameworks.
- It is easy to integrate with the testing ecosystem.
- It is open for enhancement.
- It has test-driven development.
- It's useful for comprehensive testing.

The cons of Selenium

- There are a few problems while testing.
- There are issues with finding locators.
- There are limitations in browser support.
- Manual scripts are not allowed.
- The performance is slow.

4. Sahi: An automation and testing tool

Sahi is an automation and testing tool for Web applications. It is available in both open source and proprietary versions. The open source version includes record and playback on all browsers, HTML reports, suites and batch run, and parallel playback. The Pro version includes some



Figure 3: Sahi's interface